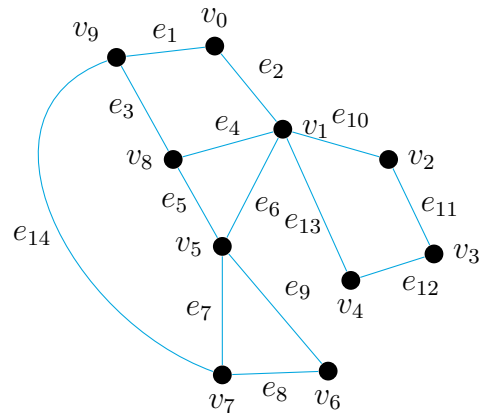


10.1.13

Determine which of the graphs in 12-17 have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.



Theorem 10.1.2

If a graph has an Euler circuit, then every vertex of the graph has positive even degree.

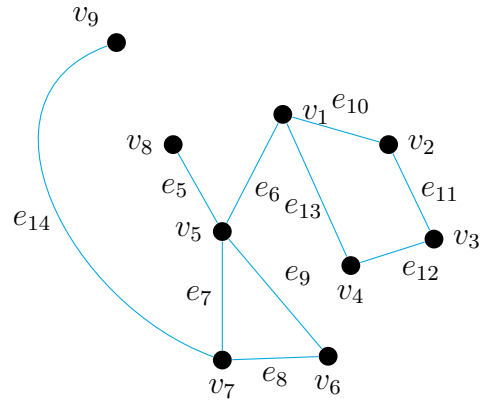
Theorem 10.1.3

If a graph G is connected and the degree of every vertex of G is a positive even integer, then G has an Euler circuit.

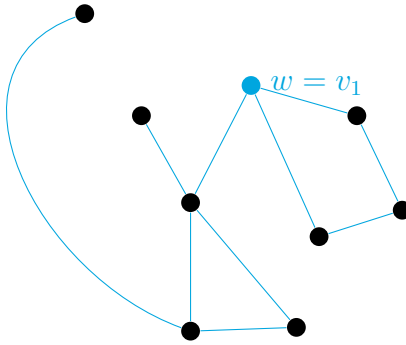
Proof

Let G be the graph. Suppose that the degree of every vertex of G is a positive even integer. [We must find an Euler circuit for G]. We can construct a circuit C by the following algorithm:

Step 1: Pick any vertex v of G at which to start: v_0 .

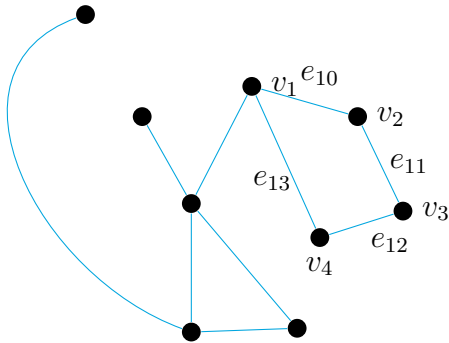


Step 3b: Pick any vertex w common to both C and G' . Choose $w = v_1$.



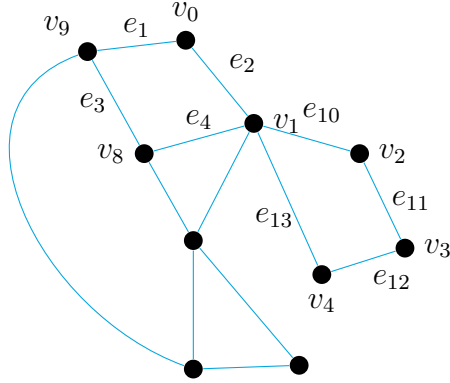
Step 3c: Pick any sequence of adjacent vertices and edges of G' , starting and ending at $w = v_1$ and never repeating an edge. Call the resulting circuit C' .

$$C' = v_1 e_{10} v_2 e_{11} v_3 e_{12} v_4 e_{13} v_1$$



Step 3d: Patch C and C' together to create a new circuit C'' . Start at v_0 and follow C to $w = v_1$, follow C' all the way back to v_1 , and then continue along the untraveled portion of C back to v_0 .

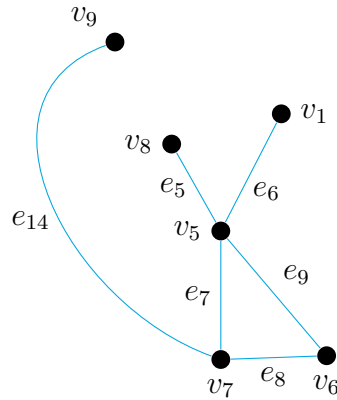
$$C'' = v_0 e_2 v_1 e_{10} v_2 e_{11} v_3 e_{12} v_4 e_{13} v_1 e_4 v_8 e_3 v_9 e_1 v_0$$



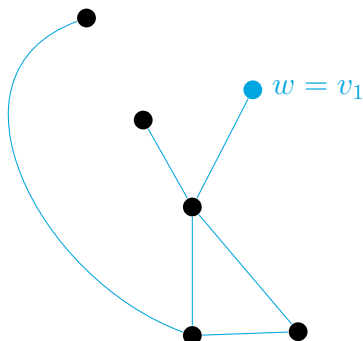
Step 3e: Let $C = C''$ and go back to Step 3.

Step 3: Check whether the new C contains every edge and vertex of G . It still does not. The edges $e_5, e_6, e_7, e_8, e_9, e_{14}$ have not been included, so we proceed with Step 3a again.

Step 3a: Remove all edges of the new C from G and remove any vertices that become isolated. Call the resulting subgraph G' .

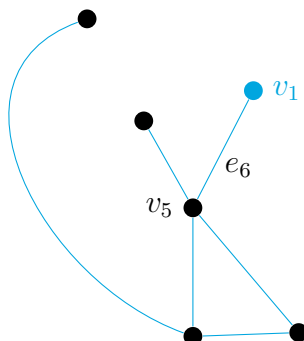


Step 3b: Pick any vertex w common to both C and G' . Choose $w = v_1$.



Step 3c: We now try to pick a sequence of adjacent vertices and edges of G' , starting and ending at $w = v_1$ and never repeating an edge.

The only edge of G' incident with v_1 is e_6 . Thus, after leaving v_1 along e_6 , there is no different unused edge by which to return to v_1 . Consequently, the circuit required by Step 3c cannot be constructed from this choice of w .



The construction cannot be completed. Each time a circuit passes through a vertex, it uses two edges incident with that vertex: one edge to enter and one edge to leave.

However, the graph contains vertices of odd degree:

$$\deg(v_1) = 5, \quad \deg(v_7) = 3, \quad \deg(v_8) = 3, \quad \deg(v_9) = 3.$$

Therefore, the edges incident with these vertices cannot all be paired as an edge entering and an edge leaving. This is why the construction of Theorem 10.1.3 cannot continue until every edge of G has been included.

By the contrapositive of Theorem 10.1.2, if some vertex of a graph has odd degree, then the graph does not have an Euler circuit.

Solution: The graph does not have an Euler circuit.